

2. Getting Started with Cloud Shell and gcloud: Discover the use of gcloud commands to manage Google Cloud resources from Cloud Shell.

Before begin with **gcloud** commands, lets execute some linux commands here and check their functionalities:

What is Cloud Shell?

Cloud Shell is a browser-based command-line environment provided by Google Cloud.

Key points

- Runs on a temporary VM (Linux)
- Comes preinstalled with:
 - **gcloud** (Google Cloud CLI)
 - **kubectl**
 - **gsutil**
 - Git, Python, Node.js, etc.
- Free to use (with limits)
- Automatically authenticated to your Google account

Simple Practice Tasks (No Subscription)

Go to google cloud console → authorize with your gmail credentials
→ search for cloud shell from search bar and select → open terminal
once cloud shell is selected → execute the followings commands

mkdir demo

```
cd demo
echo "Hello Cloud Shell" > hello.txt
cat hello.txt
chmod +x hello.txt
ls -l
```

```
cd ..
```

File & Directory Management

```
mkdir test          # Create directory
rmdir test          # Delete empty directory
rm file.txt         # Delete file
rm -r folder        # Delete folder recursively
cp a.txt b.txt      # Copy file
mv a.txt folder/    # Move or rename file
```

Viewing & Editing Files

```
cat file.txt        # View file content
less file.txt       # Scroll view
head file.txt       # First 10 lines
tail file.txt       # Last 10 lines
nano file.txt       # Simple editor
vi file.txt         # Advanced editor
```

System Information

```
whoami              # Current user
```

```
uname -a      # OS details
df -h         # Disk usage
du -sh *      # Folder sizes
free -h       # Memory usage
uptime        # System running time
```

Environment Variables & Shell

```
env           # List environment variables
echo $HOME    # Print variable
export MYVAR=123 # Set variable
history       # Command history
clear         # Clear terminal
```

Next step is exploring the gcloud commands

Basic gcloud Configuration

1. Check active account

gcloud auth list

2. Check active project

gcloud config list project

3. Get help

```
gcloud help
gcloud compute --help
```

4. List projects

```
gcloud projects list
```

Using Cloud Shell Editor

Cloud Shell also includes a **code editor**.

Open it using:

```
cloudshell edit a.txt
```

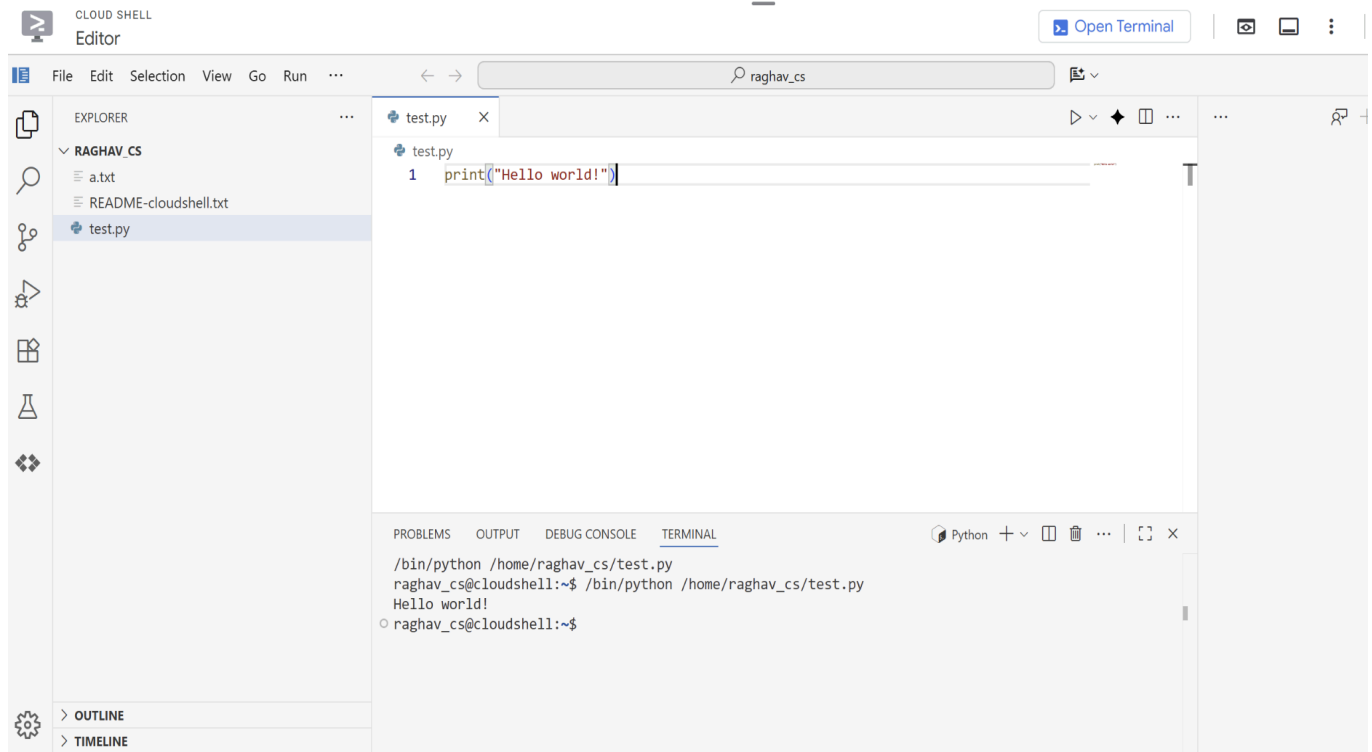
//# a file will be opened named a.txt on editor; you add text to it; go back to terminal and check its content using

```
cat a.txt
```

Note: You may execute various programming files/codes as well

For Example:

```
cloudshell edit test.py
```



From terminal you can execute the file as follows:

```
python3 test.py
```

- ◆ **Compute Engine (VMs) // Works only with Billing**

- 1. List VM instances**

```
gcloud compute instances list
```

- 2. Create a VM**

```
gcloud compute instances create my-vm \
  --zone=us-central1-a
```

- 3. Delete a VM**

```
gcloud compute instances delete my-vm \  
  --zone=us-central1-a
```

- ♦ **Cloud Storage // Works only with billing**

- 1. List buckets**

```
gcloud storage buckets list
```

- 2. Create a bucket**

```
gcloud storage buckets create gs://my-bucket-name
```

- ♦ **App Engine**

- 3. Deploy app**

```
gcloud app deploy
```

- 4. View app**

```
gcloud app browse
```

- ♦ **IAM (Permissions)**

List service accounts

```
gcloud iam service-accounts list
```
